Topological median algebra structures on ER homology manifolds II: web structures

Mladen Bestvina, Ken Bromberg and Michah Sageev

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The goal here is to work the other way around from the first paper: given a web structure, produce a median structure. Need examples. First is surface group, which has many web structure. 3-manifolds?

1 Definitions

We let M be an n-dimensional ER homology manifold. We actually don't need M to be an ER homology manifold. We could just say a locally compact, Hausdorff, path-connected and whatever else may be needed... Seems the right assumptions are Hausdorff, connected, σ-compact. (We are locally a compact CAT(0) cube complex, so don't need to say locally compact, and locally path connected, so path connected.) Do we need to say Hausdorff?

We will use the term *standard cube* in R^n to refer to $[0,1]^n$. More generally, a *box* in R^n is a product of n compact intervals. A box has a collection of n *standard codimension-1 foliations*, whose leaves (called *plaques*) are preimages of points under one of the standard projections to one of the interval factors. One could talk of plaques of arbitrary dimension or codimension which appear as intersections of codimension-1 plaques. We will generally use the term "plaque" to refer to codimension-1 plaques, and k-plaques when we want to refer to a plaque of some particular dimension or codimension-k plaques for some particular codimension.

Definition 1.1. A *local cubulation* of M is the following structure

1. For each $x \in M$, we have a compact neighborhood N_x and a homeomorphism to N_x from a CAT(0) cube complex complex which consists of finitely many cubes around a vertex, with x is identified with the central vertex of this complex. We call this a *cubulated neighborhood*

of x, and the homeomorphic images of the cubes of N_x in M are called *local cubes.*

2. Any two local cubes (in possibly different neighborhoods) in M intersect in a collection of subcubes of both (perhaps degenerate, i.e lower dimensional), and their standard foliations match up. Formally, each cube C comes with a homeomorphism $[0,1]^n \to C \subset M$, and each transition map is a homeomorphism between collections of boxes and on each box it has the form $f : \prod_n I_n \to \prod_n J_n$ satisfying

$$f(x_1, ..., x_n) = (f_1(x_{\sigma(1)}), ..., f_n(x_{\sigma(n)})),$$

where the f_i are homeomorphisms between compact intervals and $\sigma \in S_n$ is a permutation of the indices.

Lemma 1.2 (Regularity)**.** *Given a local cubulation structure on M*

1. *any two cubes intersect in finitely many cubes*
2. *there exists a locally finite atlas of cubes such that any two cubes intersected in a single cube*

Proof. Suppose that $Q = I_1 \times \cdots \times I_n$ and $Q' = J_1 \times \cdots \times J_n$ are two cubes. Every component of $Q \cap Q'$ is a cube $K_1 \times \cdots \times K_n$ with $K_i \subset I_i$ and $K_i \subset J_{\sigma(i)}$ subintervals for some permutation σ of $\{1, \cdots, n\}$. Suppose that there are infinitely many such components. Then some permutation σ appears infinitely many often. We may pass to this infinite subset, and then reorder the coordinates, so that $\sigma = id$. Thus $K_i \subset J_i$. same dimension wlog –mb maybe a word about why? –mb

We orient each I_i, J_i and then a subinterval $K_i \subset I_i$ (or $K_i \subset J_i$) can have 4 types:

- $K_i \subset int I_i$,
- $K_i = I_i$,
- K_i contains the left endpoint of I_i but not the right endpoint, and
- K_i contains the right endpoint of I_i but not the left endpoint.

After passing to a further infinite subset of components we may assume that for each of them the types of K_i in I_i and J_i are the same. Note that for at least one i the inclusion $K_i \subset I_i$ must have type 1, i.e. must be an interior interval, for otherwise all the cubes in the collection would contain a fixed corner of Q. So again permuting the coordinates we can write each intersection cube as

$$I'_1 \times \cdots \times I'_k \times I''_{k+1} \times \cdots \times I''_n$$

where $k \geq 1$, each I'_i has type 1 and each I''_i has type 2,3 or 4. Now notice that $K_i \subset J_i$ must have type 2, i.e. $K_i = J_i$, when $i \leq k$. Thus we can write intersection cubes as is that clear? –mb

$$J_1 \times \cdots \times J_k \times J'_{k+1} \times \cdots \times J'_n$$

with each J'_i having any of the 4 types.

Now consider plaques of the form $J_1 \times \cdots \times J_k \times \{p_{k+1}, \cdots, p_n\}$ in the intersection cubes. Choose sequence of distinct components where these plaques converge to a necessarily k-dimensional plaque of the same form in Q'. On the other hand, in Q, any limit will necessarily have dimension $< k$. Indeed, the projections of the intersection cubes to the first k coordinates are pairwise disjoint and the corresponding plaques have their k-volumes converging to 0. clear? –mb

For the second claim in the lemma, choose an exhaustion of M by compact sets C_i, $i \geq 0$ so that $C_0 = \emptyset$ and $C_i \subset intC_{i+1}$. Let $A_i = C_i \setminus C_{i-1}$ for $i \geq 1$. Then choose sufficiently small cubulated neighborhoods so that if one intersects A_i and A_j then $|i - j| \leq 1$. Finally, for each i pass to a finite subcollection whose interiors cover A_i. This produces a locally finite collection $N_1, N_2, \cdots$. Now inductively subdivide each N_i so that each cube of N_i intersects each cube in N_j for $j < i$ in at most one cube. This is possible by the first part of the lemma. $\square$ good enough? –mb

Note that local cubes can fit together to give larger cubes (for example, in dimension two, four squares around a point). When that happens, we refer to these larger cubes as *cubes* of the local cubulation structure as well, even though they may not be local cubes. A point $x \in M$ is called *regular* if it has a local cubulated neighborhood of the form $N_x = X \times I$. A *singular point* of M is a point which is not regular. Note that if $x \in M$ is a singular point, then there is a neighborhood of M in which x is the only singular point, so that the set of singular points is a discrete subset of M.

Generically, the images of the plaques in M fit together to become codimension-1 *regular leaves* in M (see Section 3.4 for a formal definition). At this stage, we do not yet know much about them, but we will show that they are typically embedded and have various nice properties. Note also that one can consider plaques of any dimension (preimages of points under a projection to a product of some of the interval factors) and by definition they match up as well.

A *complete local cubulation* of M is a local cubulation satisfying the *3-cube condition*: whenever three 2-cubes share a vertex so that any two share an edge, there exists a 3-cube whose boundary contains the three given 2-cubes. Note that the three 2-cubes may not be in a single cubulated neighborhood, so this is not implied by the local CAT(0) condition.

A *web structure* on M is a complete local cubulation together with the *uniqueness of limits condition*: If a sequence of regular leaves $\{\ell_i\}$ converges to a leaf ℓ, then it does not converge to any other leaf. We will define in Section **??** what we mean by convergence of leaves.

2 Disk diagrams

2.1 Disk diagrams in CAT(0) cube complexes

Before getting to disk diagrams for higher dimensional web structures, we first review the disk diagram picture for CAT(0) cube complexes. The material here is discussed in various places, for example in [**?**].

Let X be a CAT(0) cube complex. We will be interested in 1-skeleton loops and the disks that they bound. When we have such a loop, we may find a 2-disk D that is mapped to the 2-skeleton $f : D \to X^{(2)}$, so that $f_{\partial D} = \alpha$. We arrange that this map is transverse to the hyperplanes of the cube complex. This can be done, for example, by employing the simplicial approximation theorem. The result is that the pull back under f of the union of the hyperplanes of X is a union $\mathscr{A}$ of a finite collection of loops and arcs with endpoints on the ∂D, and these loops and arcs are in general position.

Now if there are complementary components that are not disks, it means that there is a component of $\mathscr{A}$ containing no arcs. When this happens that f can be adjusted so that this "island is removed". See Figure 1. Let $\mathscr{A}'$ be a component of $\mathscr{A}$ consisting only of loops. Consider a loop α which is the boundary of a regular neighborhood the union of the loops in $\mathscr{A}'$ and which separates $\mathscr{A}'$ from the boundary of D. Now α is mapped to a contractible region R contained in the star of a vertex v in X. Since R is simply connected, we can redefine f on the region D' bounded by α, so that $f(D')$ is disjoint from the hyperplanes of X. For this new map, $\mathscr{A}$ is the same except with the loops of $\mathscr{A}'$ removed.

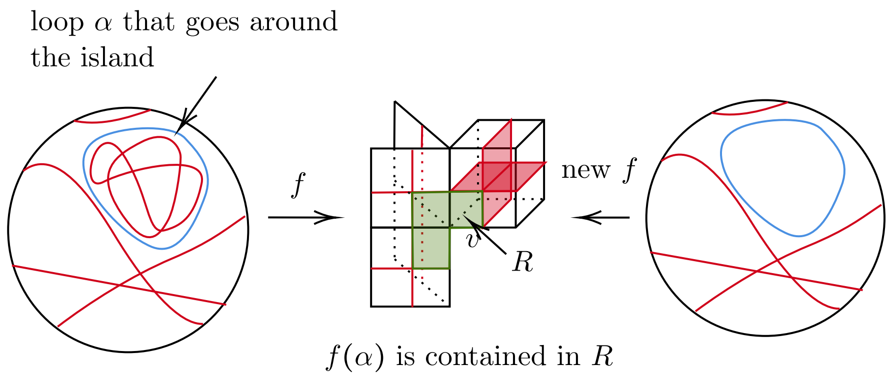


Figure 1: Removing an island: a component of $\mathscr{A}$.

To see this, note that if $\mathscr{A}$ were disconnected, there would be a complementary subsurface R with more than one boundary component. Let α be a loop in R isotopic to a boundary component of R. Since α does not cross $\mathscr{T}$, the image of α in M must be contained in the star of a vertex in some locally cubulated neighborhood. Since this neighborhood is contractible, we can redefine f on the subdisk of D' of D bounded by α, so that $f(D')$ does not meet any hyperplanes in the locally cubulated neighborhood. This eliminates one of the components of $\mathscr{T}$. Since $\mathscr{T}$ is a finite graph, after finitely many steps, we obtain a $\mathscr{T}$ which is connected. It then follows that all the complementary subsurfaces are disks.

We can arrange that this map is cellular. This means D is subdivided into squares and each square σ is mapped either to a square in X or is to to an edge by a map that factors through a projection onto one of edges of σ. If σ is mapped onto a square of X, we have that the two mid-edges of the square are mapped to hyperplanes of X. When σ is mapped to an edge of X, we have one mid-edge of σ that is mapped to a hyperplane of X. Putting together the collection of these mid-edges of squares of D together, we get a finite collection $\mathscr{A}$ of curves and arcs in D that intersect transversely. We define the area of D to be $area(D) = (n, m)$, where $n = |\mathscr{A}|$ and m is the number of intersections points of elements of $\mathscr{A}$. We order area lexicographically so that any given 1-skeleton loop has a minimal area disk diagram.

Since X is CAT(0) we do have the following local rules in $\mathscr{A}$ (see Figure 2).

1. There does not exist a simple closed loop α not crossing any other elements of $\mathscr{A}$. This would mean the entire loop is sent to a hyperplane

and not crossing any other hyperplane. We could then redefine the map f so that α is not there.

2. There does not exist a monogon. That is, an arc which crosses itself transversely once and thus bounding a subdisk E of D, with no other arcs of A intersecting E. This would mean that in X, there is a square which has neighboring edges identified.

3. There are no local bigons. That is, two arcs which intersect in a pair of points bounding a subdisk E and no other arcs of A intersecting E. Since no two squares of X intersect in a pair of adjacent edges, the endpoints of such a bigon are mapped to the same point. Thus the bigon can be homotoped locally to make the arcs disjoint. It folows that that minimal area disk diagrams have no such local bigons.

4. One can do triangle moves. If there exists a cycle of 3 arcs that bound a disk E such that no other arcs of A intersect E, then we can move one of the arcs across the other two, as in the third Reidemeister move.

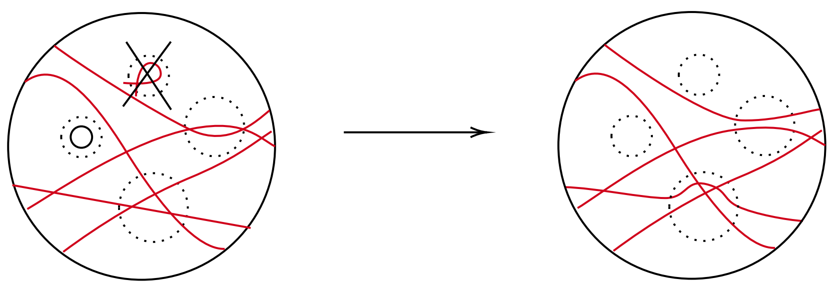


Figure 2: The elementary moves.

Now we can use local rules to show that minimal area disk diagrams have nice properties.

Lemma 2.1. *For a minimal area disk diagram, the collection $\mathscr{A}$ consists of embedded arcs that intersect at most once.*

Proof. This is done by applying the above moves. $\square$

Now we want to give the analogue of Greendlingers lemma in this context. For the following, refer to Figure 3. The closure of a complementary region of $\mathscr{A}$ can be viewed as a polygon whose edges are subarcs of arc in $\mathscr{A}$ and the boundary of D, we call these the *regions* of the diagram. A triangular region formed by two intersecting arcs and a subarc of the boundary arc is called a *corner*. (This is because in the non-dualized diagram, this corresponds to a corner of a square in the boundary.) A subdisk of D which is formed by two intersecting arcs and a piece of the boundary, and for which one of the bounding internal arcs of the subdisk does not meet any other arcs of $\mathscr{A}$ is called a *half-corner*. A boundary region which is bounded by a single element of $\mathscr{A}$ and a boundary arc is called a *spike*. (This is because in the non-dualized picture we see a spike in the boundary loop.)

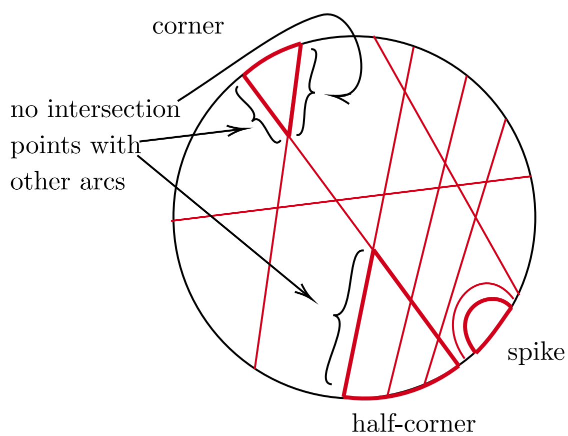


Figure 3: Corners, half-corners and spikes.

We claim the following

Lemma 2.2. *If D is a disk diagram, then either*

1. *area$(D) = (n, 0)$, for any three arcs, one separates the other two and D has two spikes.*

2. *D has at least 3 spikes or corners.*

Proof. We consider first the case that the union of the arcs in $\mathscr{A}$ is connected. If $\mathscr{A}$ consists of a single arc, we have two spikes and we are done, so let us assume that $|\mathscr{A}| > 1$. Since there cannot be spikes in this case, we will show that there are 3 corners.

We first claim that every half-corner contains a corner. Namely, for every half-corner R, there exists a corner C with $C \subset R$. We refer to Figure 4 for this argument.

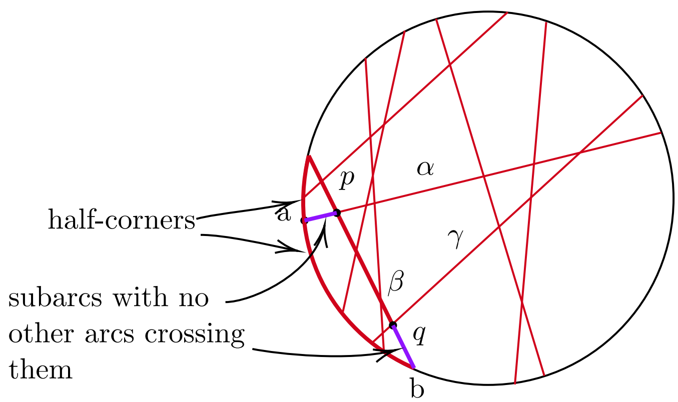


Figure 4: Finding a half corner in a half corner.

We first claim that every half-corner that is not a corner properly contains another half-corner. To see this, we first fix some notation. Let p be an intersection point of two arcs $\alpha, \beta \in \mathscr{A}$, and let a be an endpoint of α and b and endpoint of β. We let α_{pa} denote the subarc of α bounded by p and a, and similarly, we let β_{pb} denote the subarc of β bounded by p and b. Suppose that α_{pa} and β_{pb} bound a half corner, so that (without loss of generality) α_{pa} does not contain any intersection points of arcs of $\mathscr{A}$. Consider the intersection point $q \in \beta_{pb}$ that is closest to b. Let $\gamma \in \mathscr{A}$ be the arc of D crossing β at q. bounded by γ, β and ∂D that is contained in R. This is a half corner properly contained in R. Now since $\mathscr{A}$ is finite, every nested sequence of half corners terminates, so that the final half-corner in a nested sequence of half-corners is actually a corner.

Now we show D has 3-corners. Consider some $\alpha \in \mathscr{A}$, and let a and b be the two endpoints of α. Let $\beta \in \mathscr{A}$ be the arc crossing α at a point p closest to a and let γ be the arc crossing α at some point q closest to b. As before, let α_{pa} and α_{pb} denote the subarcs of α between a and p and b and p respectively. Note that β forms two half corners R_1 and R_2 with α_{pa}, so that by the above each of R_1 and R_2 contain a corner. This gives us two corners.

As above, we obtain two half corners which have α_{pb} as an edge. Since γ crosses β in at most one point, we then have that one of these corners is disjoint from R_1 and R_2. We then choose R_3 to be that corner and by above we obtain a corner inside R_3, producing three corners, as required.

Finally we consider the general case in which the union of the arcs in $\mathscr{A}$ is not connected. Let $\mathscr{C}$ denote the connected components of $\bigcup_{\alpha \in \mathscr{A}} \alpha$. To each element $c \in \mathscr{C}$, we may associated a disk diagram D_c, and D is obtained as the union of the D_c's by gluing along subarcs of the boundaries of the D_c's. Note that these subarcs of the boundary also do not meet any arcs in $\mathscr{A}$. Furthermore, since each arc in $\mathscr{A}$ separates D, this description of D as a finite graph of disks is actually a tree of disks, so that unless there is only one component (which has already been covered), there are *terminal D_c's*: a disk D_c that attach to the union of the other D_c's along a single arc in its boundary (see Figure 5). Unless c is a single arc, D_c produces at least two spikes or corners on the boundary. From this, we see that we get at least three spikes or corners all in all, unless it is a particularly degenerate case, as shown, when $\mathscr{A}$ consists of parallel disjoint arcs. $\square$

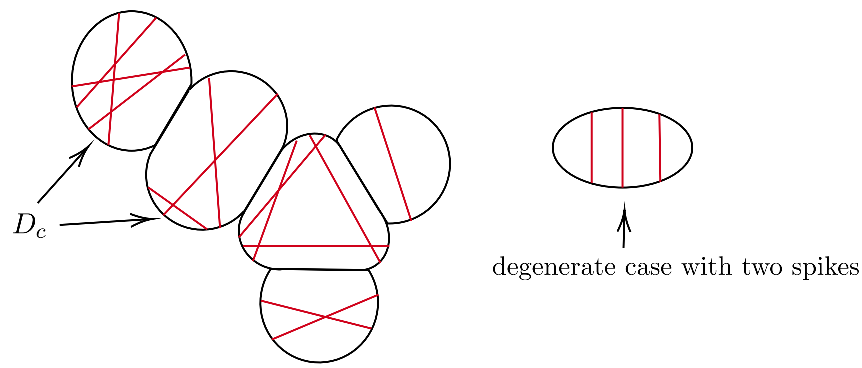


Figure 5

Dual to $\mathscr{A}$ is a van Kampen diagram V, which we describe as follows. To obtain V^0, we put a vertex in the interior of each complementary region of $\mathscr{A} \cup \partial D$. As we saw, each such complementary region is a disk. To obtain V^1, we join two vertices of V^0 by an edge when the corresponding regions share a subarc of an arc in $\mathscr{A}$ in their closure. We glue in a square whenever there is a 4-cycle running around a crossing point of two arcs. Note that since each region is on a unique side of each arc in $\mathscr{A}$ and pairs of arcs cross at most once, the closure of a complementary region intersects an arc of $\mathscr{A}$ in at most one subarc and any two complementary regions share at most at most one subarc of an arc in $\mathscr{A}$ in their closure. This tells us that V^1 is a simplicial graph. Do we want more details here? –ms

The complex V is a planar multi-disk (a contractible union of cellulated disks and arcs such as that any two meet along a boundary vertex of each), naturally embedded in D. Moreover V has the structure of a square complex Note that one can obtain D from V by attaching an annulus A to V along one of the boundary components of A and extending the hyperplanes across A disjointly. We call the interior of A the *interstitial region* (see Figure 6.) Note that the 1-skeleton path read along the boundary of D is the same as the path read along the boundary of V, so that we can further cellulate the interstitial region of D, by joining the vertices along D to the corresponding vertices in V by an embedded collection of disjoint arcs disjoint from the arcs in $\mathscr{A}$. The 2-cells of the interstitial region are called *interstitial rectangles*.

The union of the cellulated disks of V is called the *thick part* of V and the arcs that are not in the thick part are called the *thin part* of V.

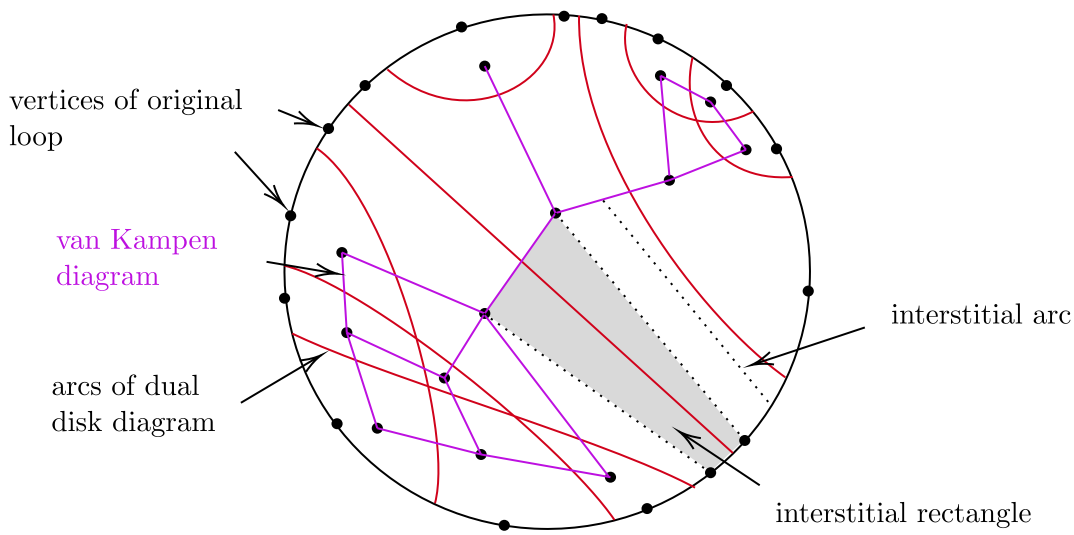


Figure 6: The disk diagram and the dual van Kampen diagram

We can now arrange the map f so that it has the following further properties:

1. $f_V : V \to X$ is a cellular map sending squares to squares and edge to edges.

2. $f : D \to X$ factors through a deformation retraction $D \to V$ collapsing each interstitial rectangle to an edge on the boundary of V

Point preimages of the map $f : D \to V$ in the interstitial regions are called *interstitial arcs*. Since we will later be talking about somewhat more complicated disk diagrams, we call the above a *standard disk diagram*.

2.2 Disk diagrams in M

We now wish to understand disk diagrams for loops in a locally cubulated space. The aim will be Theorem 2.1, which tells us that in fact for completely locally cubulated spaces, piecewise straight loops (defined below) bound the same type of disk diagram as described for CAT(0) cube complexes above. For most of this section, we will not need to assume that M is completely locally cubulated, just locally cubulated.

A path α is called *piecewise straight* if it has a subdivision into 1-dimensional cubes local. Note that since any two points in a cube of M may be joined by a piecewise straight path, any path can be perturbed rel endpoints to a piecewise straight one arbitrary close to the original path.

Given a piecewise straight loop $\alpha : S^1 \to M$, we can extend it to a map $f : D^2 \to M$ (where $S^1 = \partial D^2$). We wish to homotop f to be "piecewise straight" rel ∂D^2 as well. We do this in the following way:

- We pull back the cover of M by the interior of cubulated neighborhoods, to an open cover $\mathcal{U}$ of D.
- We choose a partition of D into polygons such that each polygon $D_i \subset U_i$ for some $U_i \in \mathcal{U}$.
- For each boundary edge e of each D_i which is not in the boundary of D, we perturb f on e relative to ∂e to make it piecewise straight. We then obtain a map f such that for each D_i, the restriction $f_{\partial D_i}$ is a piecewise straight loop in a cubulated neighborhood N_i of M.
- We subdivide N_i, so that $f(\partial D_i)$ is a 1-skeleton path in N_i and so that whenever two cells D_i and D_j intersect along an edge, the image of $D_i \cap D_j$ is a 1-skeleton path of both N_i and N_j.
- The classical disk diagram theory discussed in the previous section allows us to extend f over each D_i to a map $f : D \to M$ where D_i is a standard disk diagram and the target of each D_i is a cubulated neighborhood. Thus f_{D_i} satisfies the properties delineated in the previous section. In particular, each D_i contains a van Kampen diagram V_i, f restricted to each V_i is cellular, and f factors through a deformation retraction of D_i to V_i.

Remark. Note that while for two neighboring cells D_i and D_j above, there is a cellular decomposition of each ∂D_i as a piecewise straight path associated to the cellular structure described above, this decomposition may not agree where D_i is glued to D_j (see Figure 7). However, when there is a corner of V_i, the vertex associated to it in ∂D_i must also be associated to a vertex in V_j. This is because a straight path in M cannot also run along the corner of a square.

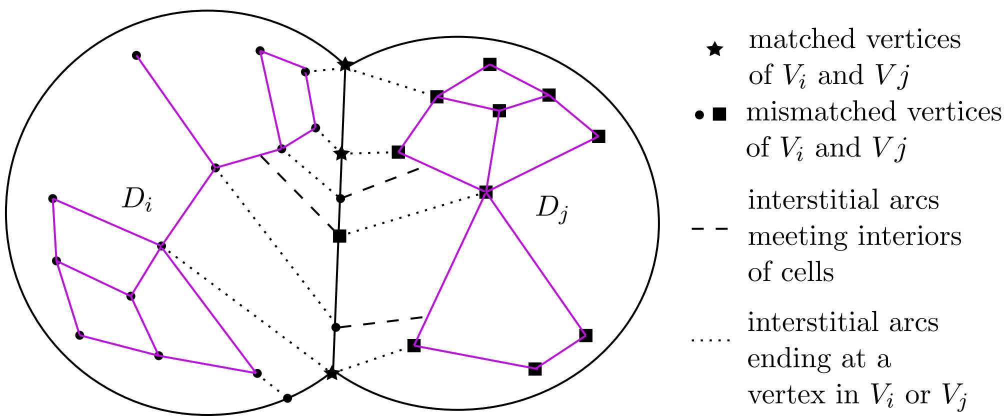


Figure 7: The squarings of two cells may not match up: points on the boundary of D_i associated to vertices in V_i may not be be associated to vertices in V_j (labeled as circles and squares here). Corners of squares in V_i do need to be paired with a vertex in V_j. These are the starred points.

We denote by V the union of all the van Kampen diagrams V_i, the complement of which is the interstitial region of the diagram.

Since there may be mismatching, we do not obtain a single disk diagram and collection of properly embedded dual arcs as in the standard case. A priori, this cannot be fixed by subdividing, as in Figure 8. Our main goal in this section is to show that in fact, a loop in M has a standard disk diagram without such pathologies.

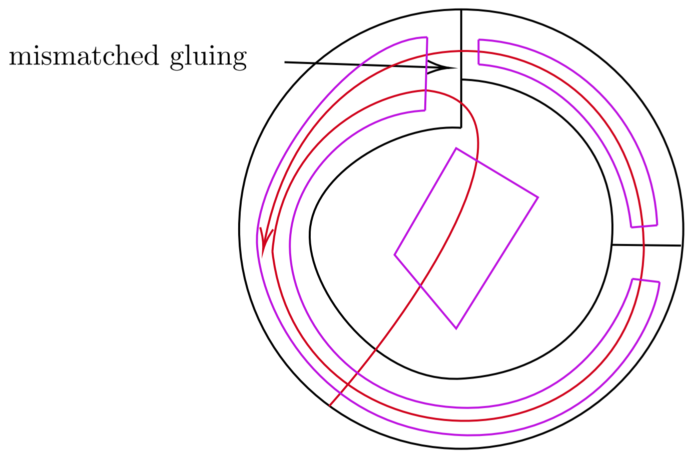


Figure 8: A possible disk diagram with infinite dual leaves. We will need to rule this out in our setting.

2.3 Train tracks

Even though the diagrams may be mismatched in our disk diagram, so that we cannot extend dual hyperplanes of squares in neighboring V_i's to match up, we can build an immersed train track that locally carries the pulled back leaves. We now explain this in more detail.

By a *leaf* of the disk diagram, we mean an immersed arc (possibly infinite) in D whose intersection with each cell in V is a union of (possibly empty) plaques and whose intersection with the interstitial regions is a union of interstitial arcs. A leaf may have an end point at a vertex of V in which we call it a *singular leaf*. Otherwise, we say it is a *regular leaf*, and its endpoints (when it is not closed) are on the boundary of D.

To construct the train track, we start by considering the standard hyperplanes in the squares and edges of V. Then we extend these slightly into the interstitial regions (see Figure 9). Suppose p is an endpoint of one such extended hyperplane, transverse to a boundary edge e in V_i. Let $q_1, \ldots, q_k$ be the endpoints of extended hyperplanes transverse to edges $e_1, \ldots, e_k$ in some other V_j, such that there exists a leaf going through the relevant interstitial regions from e to e_i. In such a situation, we put a switch point at p and draw train paths from p to each of the q_i's.

More precisely, consider two cells D_i and D_j in D which meet along an arc α (refer to Figure 9 for this discussion). We consider the edges e_i in V_j such that there exists a leaf joining a regular plaque of e and a regular plaque of e_i. When this happens, we make a legal path from p to the switch point p_i associated to e_i. Now we connect the endpoints of the arcs of the dual disk diagram in D_i if there exists a leaf running through the corresponding cells. Thus a legal path traces out the path of a leaf in D.

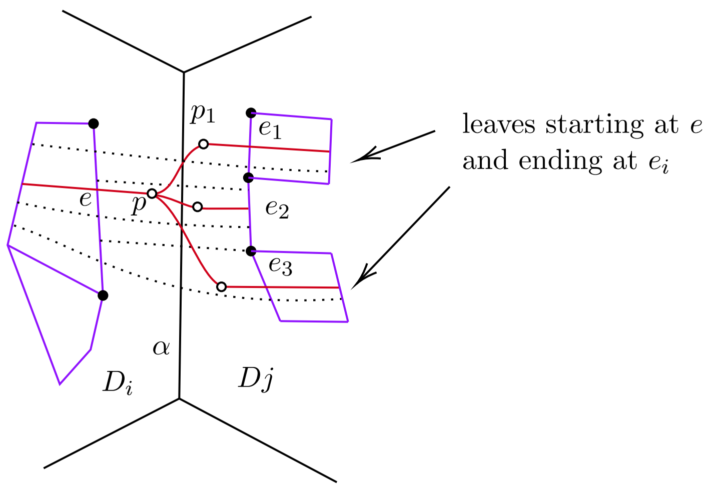


Figure 9: Building the train tracks. The track leaving the edge is given a switch point so that it can locally carry the leaves between the corresponding edges.

The union of all of these arcs is the train track, denoted $\mathscr{T}$. See Figure 10 for an example of a typical diagram. Following the standard terminology, a path in $\mathscr{T}$ has a *legal turn* at p if and only if it is smooth at p. All other local paths through p are called *illegal*. A leaf in the diagram then gives rise to a legal train path in $\mathscr{T}$.

Expand the description –mb

Better? –ms

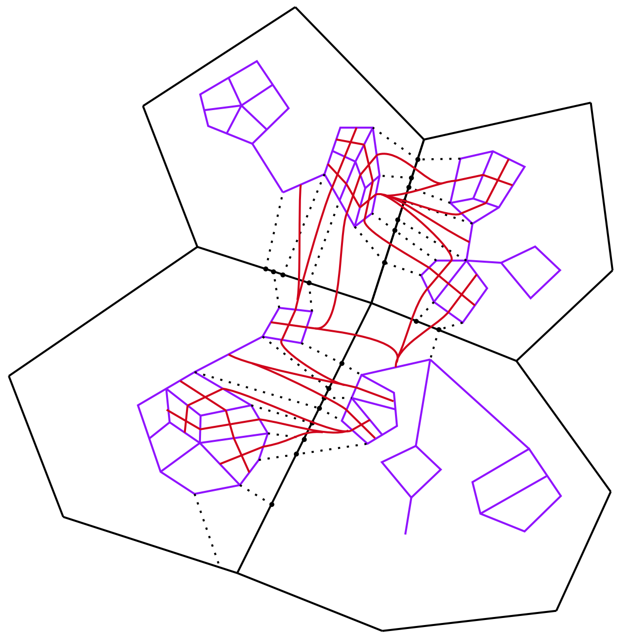


Figure 10: Building the train tracks.

Besides have switch points formed as above, $\mathscr{T}$ may have *crossing points* where the tracks cross transversely in the thick part of the diagram. We can view $\mathscr{T}$ as a graph whose vertices are the union of the crossing points and the switch points. The closure of a complementary component of $\mathscr{T}$ is called a *complementary subsurface*. An n-gon is a disk complementary component whose boundary is a concatenation of n train paths (note that n does not count switch points on the boundary of the component where the illegal turns are outside the disk).

2.4 Sliding, cross-sliding and splitting

There will be various moves we can make on tracks, which will allow us to change the diagram and ultimately improve it. To start with, the simplest move is called a *slide*, whereby we open up a switch to several switches with smaller valence (see Figure 11). The track before and after a slide would have the same legal paths and hence locally carry the same leaves. We can thus assume that $\mathscr{T}$ is generic in that all switches have valence three.

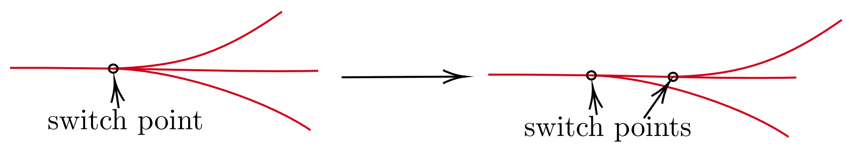


Figure 11: Sliding to make the track more generic.

In this generic situation, a switch has one arc coming in called a *large half branch* and two arcs coming out, called *small half-branches*. Sometimes, one has two adjacent half branches, as in Figure 12. The union of the two such large half-branches is called a *large branch*.

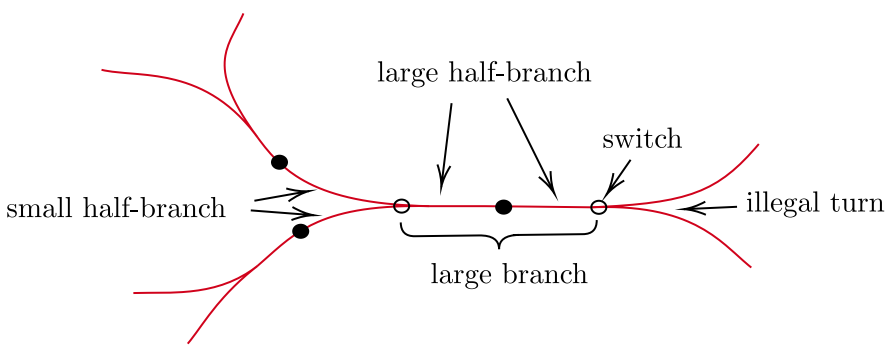


Figure 12: Terminology for arcs around a switch.

Carrying out the reverse procedure of the slide in Figure 11, requires possible subdividing, since the move may not necessarily be happening locally. See Figure 13. Here we have a small half branch at p and a large half branch at q joined together by a path α perhaps running through several thin parts of V. We then subdivide the edges crossed by α, as shown and build the track for it.

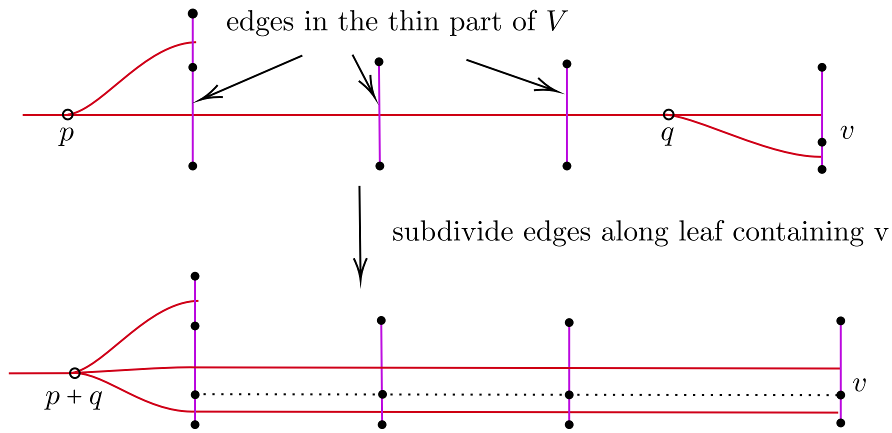


Figure 13: Sliding in reverse.

Note that putting two of these slide moves together, we get what is traditionally called sliding, as in Figure 14.

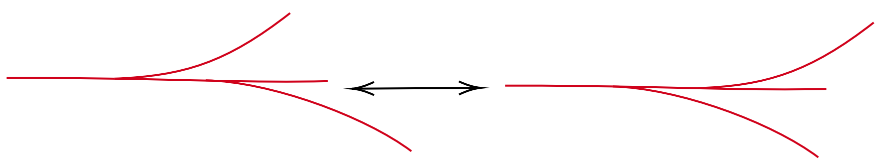


Figure 14: A traditional sliding of a switch point past another.

Another type of slide, is sliding a switch point past a crossing point. We will call this *cross-sliding.* To justify this move, we need to change our disk diagram a bit. (See Figure 15.) We split the vK diagram containing the square where the tracks cross to create a new interstitial region and then subdivide more so that the splitting happens on the other side of the intersection square.

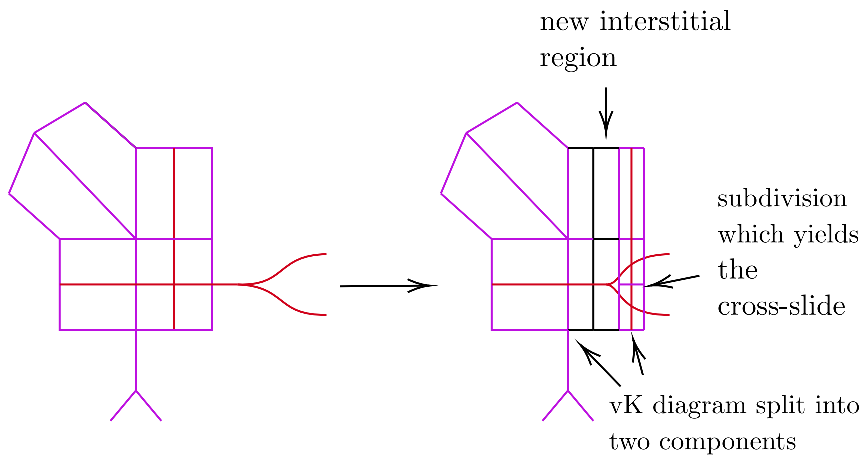


Figure 15

Another move is called *splitting*, which happens when there is a large branch. In this case, we can subdivide to eliminate the large branch and obtain two parallel arcs (called a *middle splitting*), or two other configurations, which we call *left* and *right* splittings. A middle and a right splitting are shown in Figure 16 and Figure 17 accordingly.

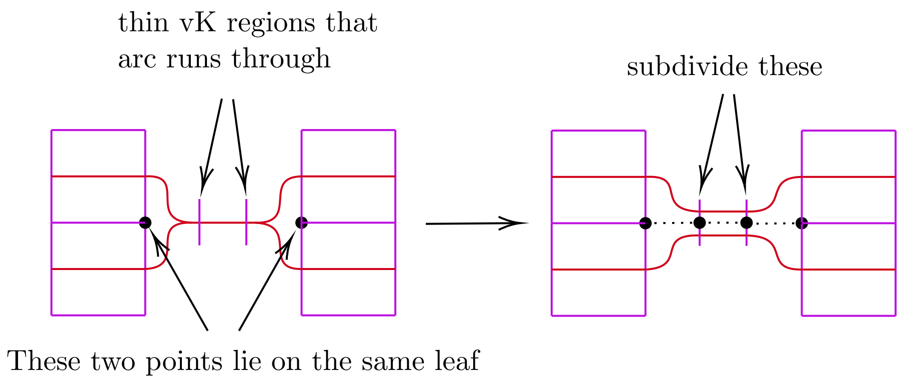


Figure 16: A middle splitting.

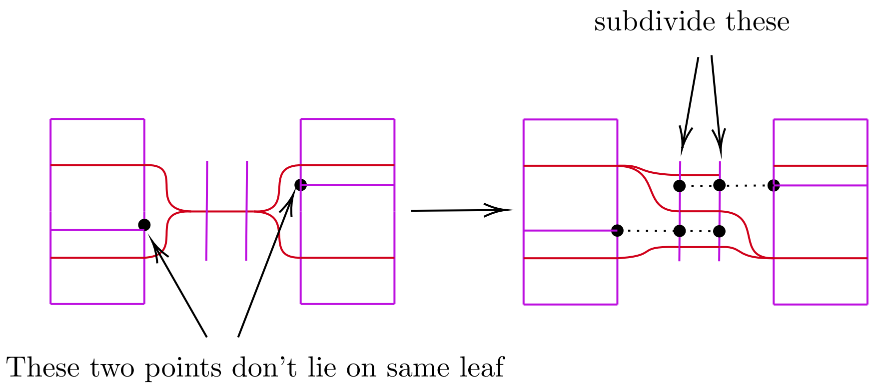


Figure 17: A right splitting.

Remark. Note that by cross-sliding, we may perform a split along a big branch even if there are crossing points along along it.

2.5 Low-gons

In this section, we want to show that there always exists a disk diagram without 0-gons, 1-gons and bigons.

Recall that the complementary regions of the immersed train tracks τ are called n-gons, where n counts the number of intersection points or switch points with illegal turn facing the interior of the n-gon. These are called the *vertices* of the n-gon. A *low-gon* is an n-gon with $n = 0, 1$ or 2. Figure 18 shows some low-gons.

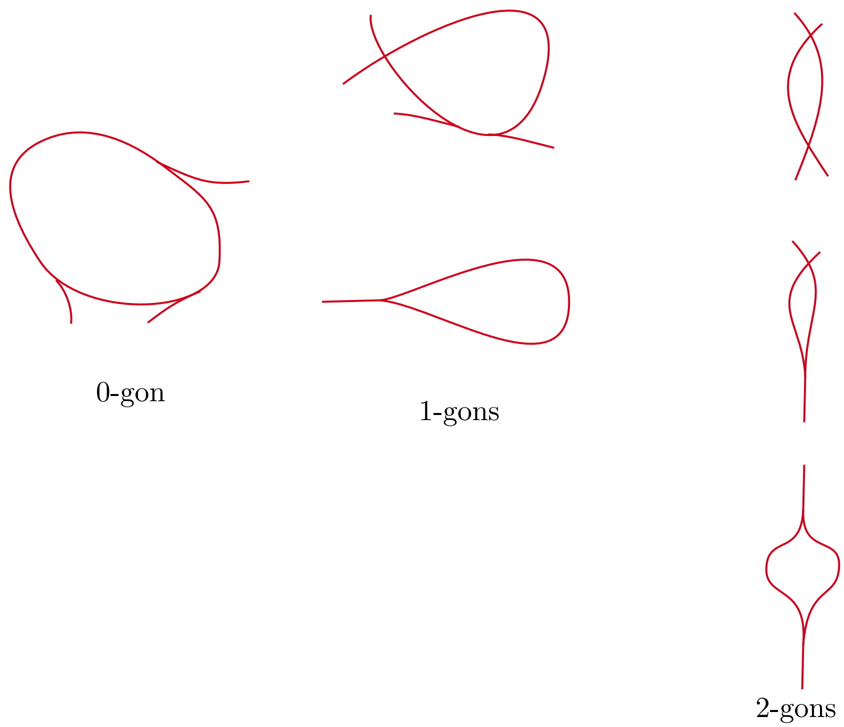


Figure 18: Some low gons.

In the above, there are n-gons (like the 0-gon), where there are switch points on the boundary on the n-gon. When they do not have such (like in the 2-gons), the n-gon is called *clean*. We have the following lemma.

Lemma 2.3. *n-gons can be cleaned: given an n-gon, we can perform a sequence of slides and splits to make the n-gon clean.*

Proof. For $n > 0$, we consider an arc A which is a piece of the boundary of on n-gon R, between vertices v and w (in the case of a 1-gon, we have $v = w$). If there is a large branch along A, we split it. Since the small half-branches that are not along A are on the same side of A, the split simply removes the large branch from A, as in Figure 19.

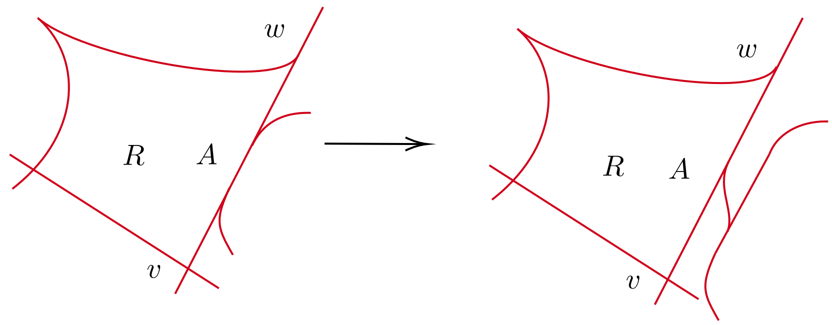


Figure 19: Splitting large branches from the boundary of the n-gon.

After removing the large branches, the switches along A are oriented as on the left in Figure 20. We can then do slides and cross slides to move all the switches off the boundary of R, as seen on the right in Figure 20.

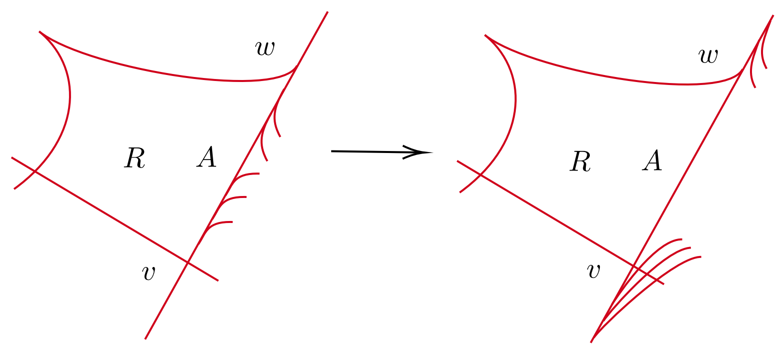


Figure 20: The boundary of the n-gon with no large branches.

When n is a 0-gon, the above process may lead to a single switch along the boundary, but such a configuration, would mean that two points along a straight path in M are identified, which is impossible, see Figure 21

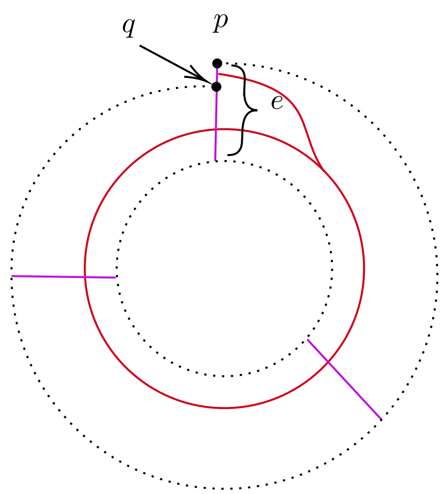


Figure 21: A 0-gon with a single switch. The edge e is embedded in M, but the points p and q are idenitified, which is impossible.

□

We then prove the following lemma addressing low-gons.

Lemma 2.4. *Given a disk diagram in a simply connected locally cubulated space M*

1. *0-gons can be excised. That is, a minimal area disk diagram does not have any 0-gons*
2. *1-gons do not exist (corresponds to gluing two adjacent sides of a square in M)*
3. *bigons can be removed using a bigon move*
4. *If M is completely locally cubulated, one can apply a triangle move to a 3-gon, as in disk diagrams for CAT(0) cube complexes*

Proof. For a 0-gon R, note first that by Lemma 2.3, we can arrange that R has no switch points on its boundary. This means that the vK edges ∂R crosses are all identified to the same straight arc c in M and there is a tubular neighborhood of $N(\partial R) \cong S^1 \times [0,1]$, so that the map $N \to M$ factors through the projection map $S^1 \times [0,1] \to [0,1]$. We can now redefine $f : D \to M$, so that ∂R is removed as a component of τ.

For a 1-gon R, first note that there are two cases. Either the vertex of the 1-gon is a crossing point, or it is a switch point with its two small half-branches along ∂R. As above, by Lemma 2.3, we first arrange that R has no other switch point on its boundary. In the case of that the vertex of the n-gon is a crossing point, we have a square of the vK-diagram being mapped to M so that two adjacent edges are identified, which is impossible. The case that the vertex of the n-gon is a switch point, we obtain two points along a single edge in one of the vK-diagrams with points on it identified, which is also impossible. See Figure 22

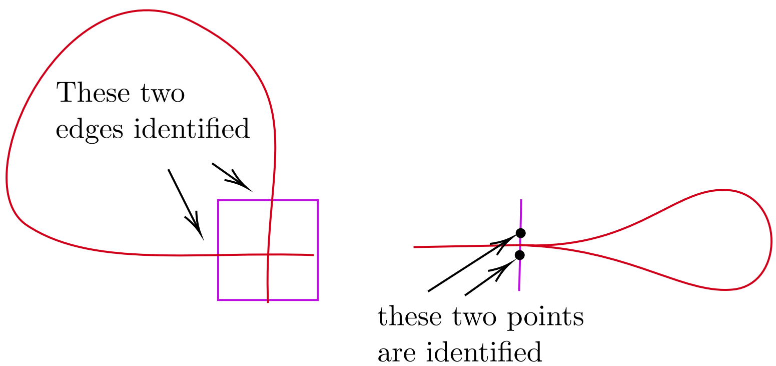


Figure 22: 1-gons are impossible.

For a bigon, we also have cases, depending on whether or not the vertices of the bigon are crossings or switches. The first is the traditional one, where the vertices of the bigon are both crossings. In the disk diagram picture, this corresponds to two squares in the diagram which are mapped to the same square in M, as in Figure 23. We can thus remove the squares and two of the crossings.

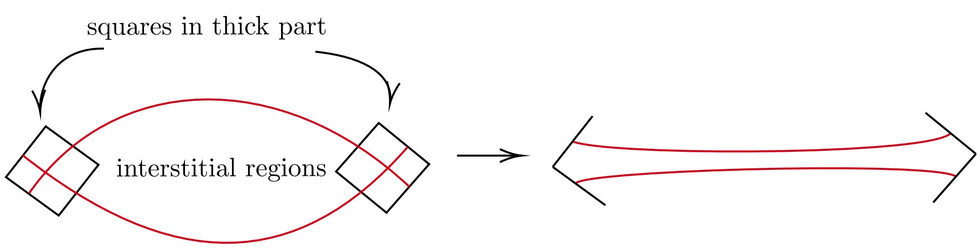


Figure 23: A bigon move removes two crossings.

Another case, is when both vertices of the bigon are switch points. The picture in the disk diagram is shown in Figure 24. Boths switches are eliminated.

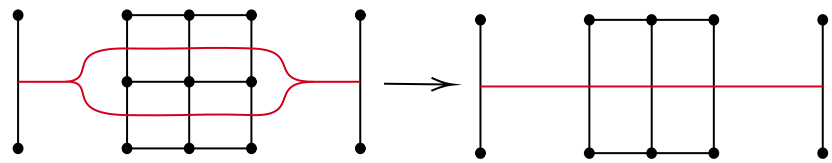


Figure 24: A bigon move eliminating two switches.

Finally, there is the case of one crossing vertex and one switch vertex. This is similarly ruled out.

Finally, we have the triangle move. By a triangle, we mean a 3-gon all of whose vertices are crossings. As in the above argument for the bigon move, we first clean the boundary of the 3-gon. This then corresponds to the disk diagram shown in Figure 25. And the triangle move corresponds to this move in the diagram.

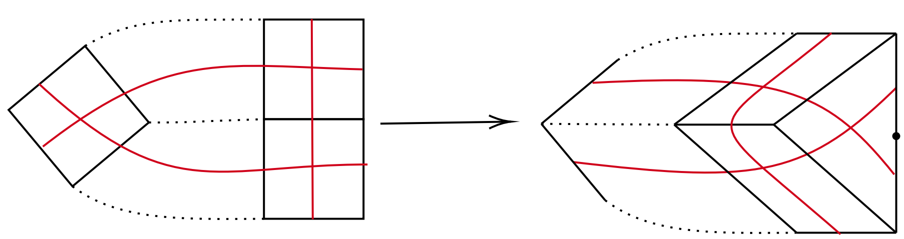


Figure 25: A triangle move.

□

Remark. Note that in the past part of the proof above, the original squares that we want to apply may be local squares in different cubulated neighborhoods and their replacement squares in this are not necessarily local squares at all. This is precisely where we need the 3-cube assumption.

2.6 Eliminating 0-loops and 1-loops

An *n-loop* is just like an n-gon, except that the region bounded by the n-loop is allowed to have non-trivial intersection with other train tracks in τ.

Lemma 2.5. *There are no 0-loops and 1-loops.*

Proof. Amongst all the 0-loops and 1-loops, we consider the innermost one α. It bounds a region R, crossed by sub-arcs of τ. Note that since α is innermost, all legal paths in R are now embedded arcs with endpoints on ∂R. We let n denote the number of legal paths in R. If there are is a large branch along a path in R, we carry out a split. This reduces n. Thus, we can carry out finitely many splits until there are no more large branches in R (see Figure 26).

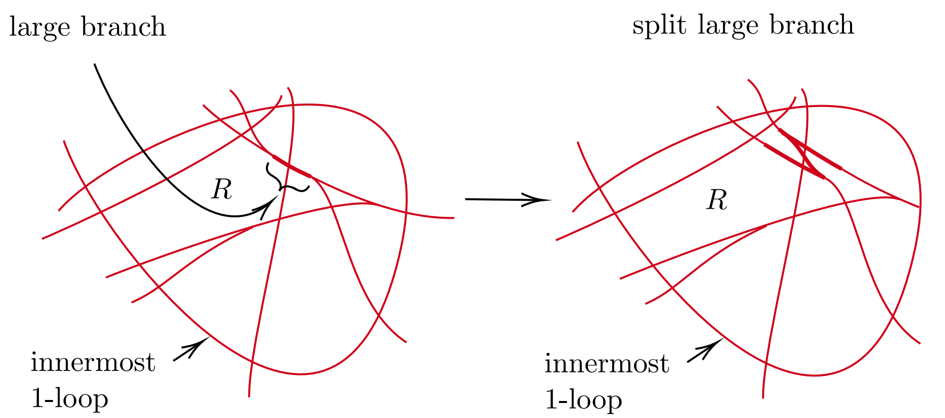


Figure 26: Splitting reduces number of legal paths in the 0-loop or 1-loop.

Now the collection of switches along a legal path in R must have the large half-branches all oriented towards the boundary, so that we can carry out slide and cross-slide moves to move them outside of R (see Figure 27.

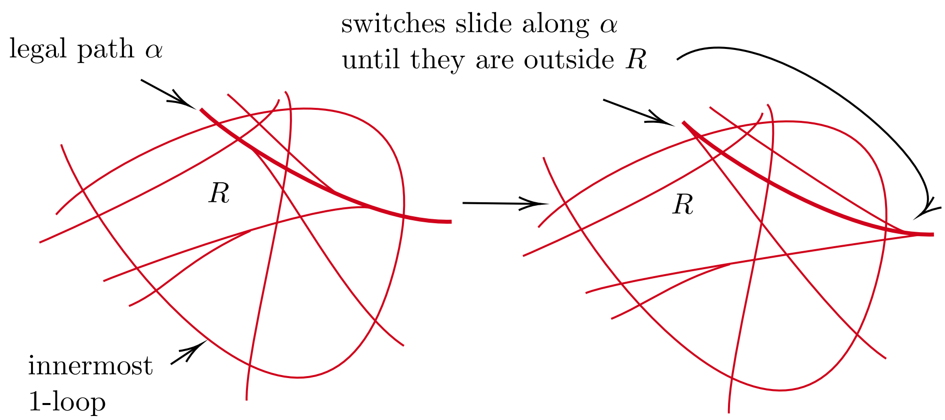


Figure 27: Sliding the switches outside of the loop.

Once we remove all the switches from the region R, we can carry out bigon and triangle moves, as in the standard disk diagram case, and obtain a 0-gon or 1-gon, which we can then rule out. □

2.7 Creating a standard disk diagram

We are now ready to to prove the main theorem of this section.

Theorem 2.1. *Every piecewise straight path in M has a standard disk diagram.*

Proof. Once 0-loops and 1-loops have been ruled out by Lemma 2.5, we know that all legal paths in D are arcs with endpoints on the boundary of D. We now proceed exactly as in the proof of Lemma 2.5 to eliminate large branches via splittings and then sliding the switches towards the boundary. The only difference is that when the switch gets to the boundary, we simply subdivide the boundary loop of D and eliminate the switch, as seen in Figure 28. □

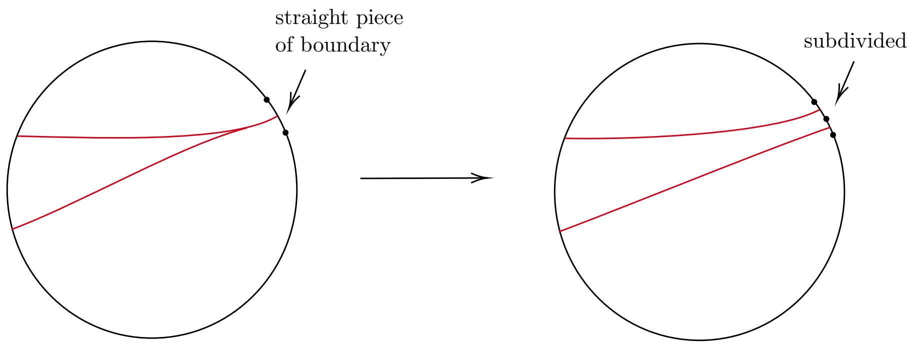


Figure 28: When the slide reaches the boundary.

3 Leaf properties

Having established the disk diagram theorems of the previous section, we will now use them to prove various properties of leaves. We assume here that M is a locally cubulated space satisfying the no missing 3-cubes property. In light of Theorem 2.1, we may assume that all disk diagrams for loops are standard and minimal.

First, we need to define regular leaves. First, we define regular leaves for a CAT(0) cube complex X. For a hyperplane H of X, let $N(H)$ denote the union of the cubes intersecting H non-trivially. Recall that $N(H)$ has a product structure of H with a unit interval. A *regular leaf* is a parallel copy of H in $int(N(H))$ for some hyperplane of H. More precisely, in the identification $N(H) \cong H \times [0,1]$, a regular leaf is a subset of the form $H \times \{t\}$ for $0 < t < 1$.

Now returning to our locally cubulated space M, a *leaf path* α joining two points x and y is a path in M with the following properties. There exists a finite collection of cubulated neighborhoods $N_1, \ldots, N_m$ and a partition of α into subpaths $\alpha = \alpha_1 \cdot \alpha_2 \cdot \ldots \cdot \alpha_m$, such that $\alpha_i \in int(N_i)$ and α_i is contained in a regular leaf of N_i. Given a plaque P in M, the *leaf* containing P is the subset of points of M that can be joined to a point in $int(P)$ by a leaf path whose first segment lies in P.

Just as for plaques, unless stated otherwise, we will use the term "leaf" for a codimension-1 leaf. A consequence of the definition is that if L is a leaf that intersects a cube of M in a plaque that is transverse to some edge e at some point p, then for all cubes that contain an edge with p in its interior, the leaf extends into those cubes as well.

A *regular leaf* is a leaf L that does not contain a singular point of M.

This means that at every point $x \in L$, there exists a 1-cube e with $x \in int(e)$ such that L meets e transversely at x . Since the collection of singular points is countable, regular leaves are generic. We will focus for now on regular leaves and define singular leaves later.

The first thing we will see is that regular leaves are embedded. In fact, we have the following.

Lemma 3.1. *Let L be a regular leaf and σ a local cube. Then $\sigma \cap L$ is a single plaque.*

Proof. Suppose that P, Q are two plaques in $\sigma \cap L$. There are two cases, depending on whether P and Q intersect or not. Let us first consider the case that P and Q intersect. In this instance, there exists a 2-face τ of σ such that $P \cap C$ and $Q \cap C$ are two intersecting 1-dimensional plaques in F. By definition, we have a leaf path joining the plaques $P \cap C$ and $Q \cap C$. This means that we have a band of squares joining an edge e parallel to $P \cap C$ and f parallel to $Q \cap C$. After truncation, we then have the two possibilities pictured in Figure 29.

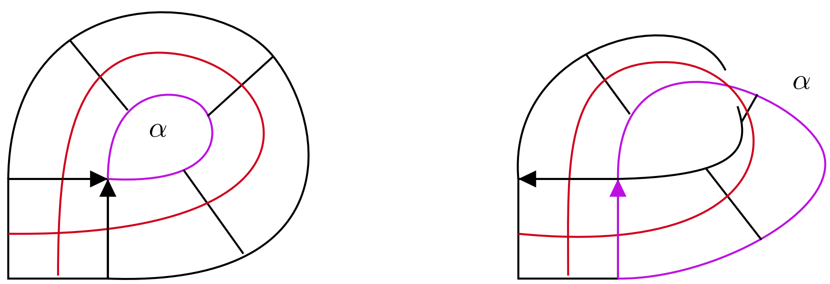


Figure 29: Two ways a regular leaf can self-intersect.

For the type I intersection, we consider a disk diagram D for the loop α. We then attach to this the band of squares between α and β to produce a disk diagram D' for β. This disk diagram D' has a 1-loop in it. Thus, as in the proof of the disk diagram theorem, after finitely many bigon and triangle moves, we can produce a disk diagram with a 1-gon, which is impossible.

For the the type II intersection, we consider a minimal disk diagram for the loop α indicated in Figure 29. Amongst all such configurations (square of self intersection, band with half twist, loop α subdivided so that it has a standard minimal disk diagram), we choose one such that the $(length(\alpha), area(D))$ is minimal. We note that by the minimality of

$length(\alpha)$, there is no backtracking along α, so that the disk D has no spikes. It follows that D has three corners. We can thus assume that there is a corner that is along some point p along α that is not an endpoint of e. Let σ_1 and σ_2 be the two squares of the band that have p as a vertex. The corner of D at p provides a square σ at p that is the image of a square σ in the vK-diagram of D. (See Figure 30.)

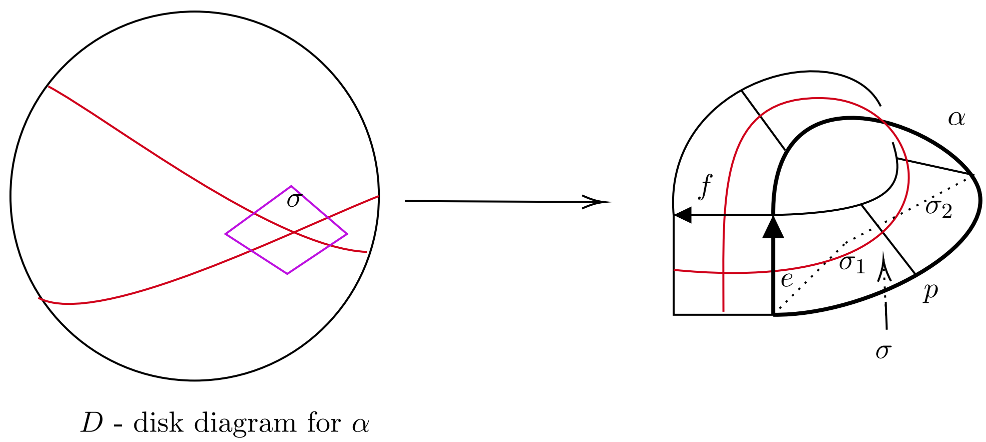
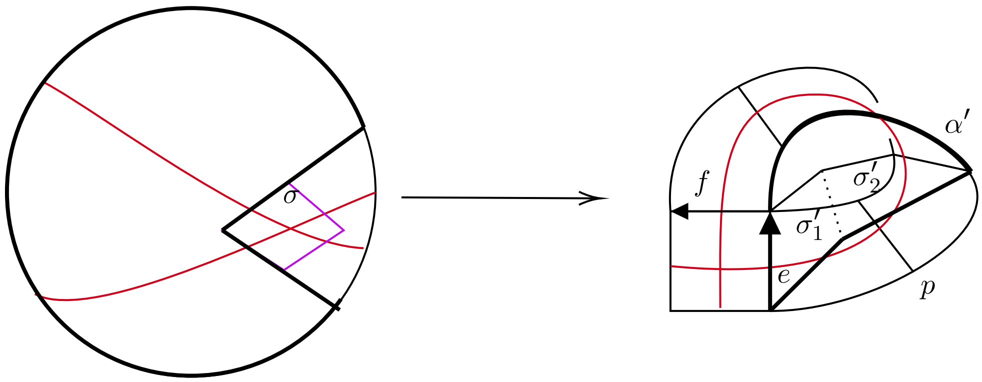


Figure 30: The corner of the disk diagram provides a new square at p.

The triple $\sigma_1, \sigma_2, \sigma$ at p tells us that there exists a 3-cube τ at p which σ_1, σ_2 and σ as faces. Let σ_1' and σ_2' be the faces opposite of σ_1 and σ_2 respectively. Replacing σ_1 and σ_2 with σ_1' and σ_2' gives us a new band and a new path α'. We also obtain a new disk D' obtained from D by cutting out the square σ in D. We thus obtain a new configuration with a disk diagram of smaller area, a contradiction. (See Figure 31.)



D' - a new disk diagram for α'

Figure 31: Replacing the band with a new band and a smaller disk diagram.

Finally we need to handle the case that the two plaques do not self intersect. This is done in a similar way as the previous case of intersecting plaques. As before, we have bands joining the endpoints of the plaques, which either give and annulus or a mobius band. In both cases, we can take a boundary loop and reduce the are of a disk bounding it by finding a corner along the boundary. □

Note that Lemma 3.1 tells us that regular leaves are embedded. Note that since the intersection of a regular leaf with a cubulated neighborhood is a hyperplane, it follows that a regular leaf itself inherits the structure of a local cubulation from the ambient space.

Together with the regularity condition, we will now see that in compact regions, a regular leaf behaves much like a hyperplane in that it has a trivially foliated regular neighborhood.

To this end, it is useful to describe a transverse holonomy on leaves. Let α be a piecewise straight path in a regular leaf L joining points a and b. We want to describe a map between an arc transverse to L at a and one transverse to L at b. To start with, if a and b are in the same plaque $P \subset L$ of an n-cube σ, then σ has a product structure of an $n-1$ cube cross a 1-cube $\sigma = H_\sigma \times I_\sigma$, where a and b have the same I_σ coordinate. (We view σ_H as the horizontal factor and I_σ as the vertical factor.) If $p : \sigma \to H_\sigma$ is the projection map, then $I_a = p^{-1}(p(a))$ is called the vertical transversal through a associated with the cube σ. We then have a natural bijection $\eta : I_a \to I_b$ which identifies points in the same plaques parallel to P. Note that if we kept the same points a and b and changed the ambient cube σ, we might get different transversals but any two such intersect in a transversal, so what is well defined is a germ of a transverse arc to a regular leaf L at a point.

Suppose now that L is a leaf which runs through the interior of two intersecting cubes σ and τ. Suppose that a and b are two points in L, one in σ and one in τ. Then the transverse arc at a in σ and the transverse arc at b in τ might not have an identification between them as above. However, we can find a subtransversal and a natural identification. More precisely, suppose that $\sigma = H_\sigma \times I_\sigma$ and $\tau = H_\tau \times I_\tau$. Then we obtain natural identifications of the transversals in $H_\sigma \times (I_\sigma \cap I_\tau)$ in the cube $H_\sigma \times (I_\sigma \cap I_\tau)$. Similary we get natural identifications between the transversal to b and to c in the cube $H_\tau \times (I_\sigma \cap I_\tau)$. The composition gives an identification between germs of transversals at a and b which identify points in the same leaves. By induction, given a leaf path α joining two points in the $a, b \in L$, we obtain a natural identification between transversals at a and b, $\eta_\alpha : I_a \to I_b$.

Now given a loop α we obtain a map $\eta_\alpha : I_a \to I'_a$ between two transversals at a. We call this the *holonomy map*.

Lemma 3.2. *The holonomy map under any leaf path α is the identity.*

Proof. Suppose that α is a leaf path and $x \in I_a$, such that $\eta_\alpha(x) \neq x$. Note that η_α identifies points in the same leaf. Thus we obtain x and $\eta_\alpha(x)$ lie in the same leaf. But $\eta_\alpha(x) \neq x$ means that x and $\eta_\alpha(x)$ are in different plaques in the same cube, a contradiction to Lemma 3.1. □

We are now ready to describe regular neighborhoods of regular leaves.

Lemma 3.3. *Let L be a regular leaf in M. Let $K \subset L$ be a finite union of cubes of M. Let $L_K = L \cap K$. Then there exists an embedding $\iota : L_K \times I \to K$ such that $\iota(L' \times \{x\})$ is a regular leaf in K.*

Proof. Let $K = \cup_{i=1}^{n} C_i$ a finite union of cubes. We may assume that L_K and K are connected, for otherwise we consider each component separately. We prove the lemma by induction on n. For $n = 1$ the claim is clear. Suppose that the lemma is true for $n-1$. Let $K' = \cup_{i=1}^{n-1} C_i$. Then we have an embedding $\iota : L_{K'} \times I \to K'$ sending horizontal regular leaves to regular leaves in K'. Let $N = \mathrm{Im}(\iota)$. We now consider the intersection of N with C_n. Let $P = L \cap C_n$, which by Lemma 3.1 is a single plaque. Now each component D_i of $N \cap C_n$ is a subcube of C_n of the form $D_j = H_j \times I_j$. By Lemma 3.2 $I_j = I_k$ for any two such components. It follows that after possibly passing to a further subinterval of I, we can extend ι from $L_{K'} \times I$ to $L_K \times I$. □ Enough details? –ms Sure –mb

We now establish further properties properties of regular leaves.

Proposition 3.4.

1. *Each regular leaf separates M into two complementary components.*
2. *Each regular leaf is simply connected.*
3. *The intersection of two regular leaves is a either empty or a regular codimension-2 leaf.*
4. *If three leaves pairwise intersect non-trivially, then the triple intersection is non-empty.*

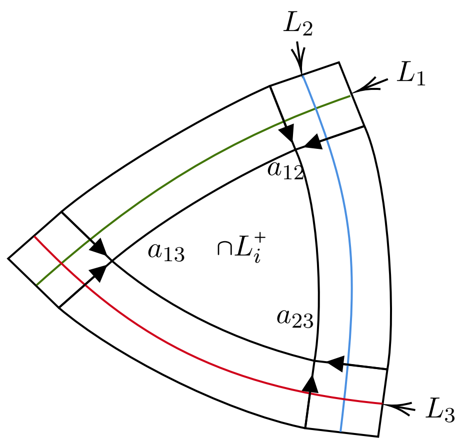
Proof.
1. To show the separation property, let L be a regular leaf and let e be a 1-plaque transverse to a plaque $P \subset L$. Let p and q denote the endpoints of e, we first show that p and q are in separate components. Suppose that there exists a path γ joining p and q in the complement of L. We homotop γ so that it is piecewise straight. The path γ together with e gives us a piecewise straight loop $\alpha = e \cup \gamma$ which crosses L only in an interior point x of e. We consider a disk diagram $f : D \to M$ for α. Since D is compact, $f(D)$ is compact. Let K be a finite union of cubes that contains $f(D)$. By Lemma 3.3, $L_K = L \cap K$ has a regular neighborhood $N(L_K)$ consisting of leaves parallel to L_K. Since γ does not meet L_K, we can choose $N(L_K)$ so that γ does not intersect $N(L_K)$ as well.

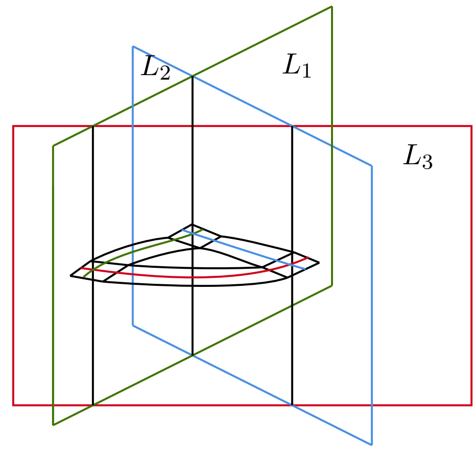
Now we consider the point x. Recall that the process of producing a disk diagram for D involves possibly subdividing α. Thus x may be a vertex of α or a point in the interior of an edge of α. In the latter case, the leaf in D associated to x lies in $f^{-1}(L)$. Since this leaf is an arc with both endpoints on α, it follows that there is a point in γ mapped to L, a contradiction. If x is a vertex of α, we note that $f^{-1}(int(N(L_K))$ is open and hence contains points in e close to x, but in the interior of an edge. Such a point is in the preimage of a leaf L' in $N(L_K)$. Again, now there is an arc of D in $f^{-1}(L')$. However, since by Lemma 3.3 $L_K = L \cap K$ has a regular neighborhood of leaves parallel to L_K, we have a leaf in D, that lies in the preimage of one such leaf L'. But since the leaves of D are arcs with endpoints on the boundary, there must be another edge in ∂D that crosses the preimage of L'. It follows that the path α crosses L at a point of γ, a contradiction.

Having shown that p and q are in different components, we now show that there are exactly two components. Given any point in $y \in M - L$, we join it by a piecewise straight path γ to L. We can choose y to be the first point of L that γ meets. So that γ meets L only at its endpoints. Now

choose a piecewise straight path β joining y and the point $x \in e$. By Lemma 3.3, we have a regular neighborhood for some compact L_K containing β. We can then push beta off of L. The endpoint of β will then be on the edge e and is on the same side as one of p or q. Thus there is a path in the complement of L from y to one of p or q.
2. To prove simple connectivity, let L be a regular leaf. Consider a loop α in L. We can homotop α in L so that it is piecewise straight. Now consider a minimal disk diagram D for α. We argue in a similar fashion to the proof of Lemma 3.1. Amongst all such α and D, with α non-trivial in $\pi_1(L)$, let $(length(\alpha), area(D))$ be minimal (in the lexicographical order). If α has backtracking, then we can reduce the length of α, so we may assume α has no backtracking. This means that D has three corners. In particular, there are two consecutive edges e and f along α that span a square σ. Since e and f are in L, so is σ. We then replace α with a path that contains the two other edges of σ. This is a path in L homotopic to α with a disk diagram of smaller area.
3. Next we turn to understanding the intersection of two regular leaves. Let L_1 and L_2 be regular leaves in M such that $L = L_1 \cap L_2 \neq 1\varnothing$. Note that the intersection of the two leaves is locally a codimension-2 leaf, so one just needs to prove that L is connected. Suppose that p and q are points lying in separate components of $L_1 \cap L_2$. For $i = 1, 2$, let $\alpha_i \in L_i$ be a piecewise straight path joining p and q. We then obtain a loop $\alpha = \alpha_1 \cdot \overline{\alpha}_2$ be the loop that is the concatenation of α_1 and the inverse of α_2. And let D be a minimal disk diagram for α. As in the previous arguments, choose p, q, α, and D such that $(length(\alpha), area(D))$ is minimal amongst all such configurations. By the 3-corner lemma, there exists a corner not at p or q along α. As above, we use this to produce a configuration of smaller area.
4. Finally, we turn to the last claim. We consider a triple of regular leaves L_1, L_2, L_3 that pairwise intersect. Assume that $L_1 \cap L_2 \cap L_3 = \varnothing$. As in the proof of Lemma 3.1, we consider points $p_{ij} \in L_i \cap L_j$ for $i \neq j \in \{1, 2, 3\}$. As the leaves are reqular, for each i, j there exist squares σ_{ij} containing $p_i j$ in their interior. As in previous arguments, we want to connect these by bands and then build an appropriate disk diagram. We need to first show that it is a standard configuration as shown in Figure **??**, and in particular the bands to not fit together to make a mobius band. To see this. As we have shown, $L_2 \cap L_3$ is connected. Thus, since it is disjoint from L_1, there exists a complementary component L_1^+ of L containing $L_2 \cap L_3$. Similarly, for L_2 and L_3, there exist complementary components L_2^+ and L_3^+ containing $L_1 \cap L_3$ and $L_1 \cap L_2$ respectively. Since $p_{ij} \in L_i \cap L_j \in L_k^+$ (for distinct i, j, k) and since our squares $\sigma_i j$ can be chosen as small as we like, we can choose σ_{ij}

such that so that $\sigma_{ij} \subset L_k^+$.





Now note that each σ_{ij} has a unique vertex $a_{ij} \in L_i^+ \cap L_j^+$. It follows that $a_{ij} \in \cap_{i=1}^3 L_i^+$. For each edge of e that contains one of the a_{ij}'s we orient the edge towards the region $\cap_{i=1}^3 L_i^+$. Now we join the edges of σ_{ij} that contain the a_{ij}'s by foliated bands. The union of these bands and the σ_{ij}'s is then an annulus with one component of its boundary, α, containing the a_{ij}'s.

Let β and the other boundary component of the annulus. Consider a minimal disk diagram D' for α. Then D' together with the annulus, is a disk diagram for β. This disk diagram has three arcs $\alpha_1, \alpha_2, \alpha_3$ that form a 3-loop in D. We then can carry out bigon and triangle moves to "clear" all the arcs that cross this 3-loop, which creates a 3-gon. This 3-gon corresponds to 3 squares in M around a vertex and L_1, L_2 and L_3 are leaves that cross the edges of these squares. By the no missing cubes condition, there exists a 3-cube of which these three squares are faces. The three leaves L_1, L_2, L_3 now meet in this 3-cube, a contradiction. □

This might need some more details –ms

As a consequence of the above, we get the following Helly type property for regular leaves.

Proposition 3.5. *Let $L_1, ..., L_n$ be a collection of regular leaves such that $L_i \cap L_j \neq \varnothing$ for every i, j. Then $\cap_i L_i \neq \varnothing$.*

Proof. We observe that a regular leaf itself inherits a web structure from M, but of smaller rank. Thus, we induct on the $rank(M)$. For $rank(M) = 1$, the leaves are points and the Proposition is clear. We assume that the claim is true for L_n. Note that for each $i < n$, $L_i' = L_i \cap L_n$ is a regular leaf in L_n. Moreover, for each $i, j < n$, by Proposition 3.4, item 4, we have that $L_i' \cap L_j' \neq \varnothing$. Thus, by induction, $\cap_{i=1}^{n-1} L_i' \neq \varnothing$. Since this intersection is a subset of L_n, we obtain that $\cap_{i=1}^n L_i \neq \varnothing$. □

4 Limit leaves

Having established some useful properties for regular leaves, we now discuss limits of regular leaves, which we will need in the construction of the median structure.

We consider a sequence (L_n) of regular leaves, whose intersection with some cube σ is a sequence of parallel plaques π_n which converge to a plaque π. The limiting plaque π may be a boundary cell of σ. We will extend π to a limiting object to which L_n will converge on compact sets.

To this end, let $K \subset M$ be a finite union of maximal cubes. Note that for each n, the intersection $L_n \cap K$ is a regular leaf and hence contains no singular points of M. Since there are finitely many singular point in K, we may choose N sufficiently large, such that for all $n, m > N$, there exist no singular points between n and m. This will mean that for all $n > N$, the leaves $L_n^K = L_n \cap K$ cross the same edges and hence intersect each maximal cube of K in parallel plaques (or the intersection with all the L_n's is empty), converging to a limiting plaque. The limit leaf in L_K is the union of these limit plaques over all the maximal cubes of K.

Now we exhaust M by compact sets $M = \cup_i K_i$, where each K_i is of the above type. Note that the definition of the limiting plaque of a cube in K_i does not change in any larger K_j. We let L be the union of these limiting plaques, and we call such a leaf a *limit leaf*. Recall that the definition of a web such a limiting leaf is unique. That is, the leaf L is connected.

Example of disconnected limit.

We claim that as for regular leaves limit leaves inherit a web structure from M.

Proposition 4.1. *Let L be a limit leaf. Then L satisfies all the propertiein-herits a web structure by intersecting the cubes of M with L.*

Proof. We must see that L has a local cubulation, that it satisfies the no-missing 3-cube condition and that leaves limit to a connected leaf. Let (L_n) be a sequence of leaves converging to L.

Local cubulation. Consider a point $x \in L$. By construction, there exists an edge e with x as an endpoint, such that every L_n is transverse to e. Note that regular leaves are locally CAT(0) cubulated since it follows from the facts that hyperplanes are CAT(0) cubulated in CAT(0) cube complexes (see [**?**]). The local cubical structure of x at L is the same as the local cubical structure of L_n at x. Thus L is locally CAT(0) cubulated as required.

No missing 3-cubes. If a triple of edges in L which span a 3-cube σ, then σ is also contained in L. Thus a triple of squares with a missing 3-cube in L, would give such a configuration in M, a contradiction.

might want more details here –ms

Connected limit. First suppose that a sequence of leaves L_n in L converge to disconnected L'. the sequence L_n

□

Remark. Note that a limit leaf may not separate M. For example consider the upper Euclidean halfplane squared in the standard way and a sequence of leaves of leaves $L_n = \{(x, 1/n) | x \in R\}$. Then the limit leaf is the x-axis which does not separate.

5 Constructing the median structure

The construction here follows what was done in dimension 2. Let M be an ER homology manifold witht a web structure. We define $\mathscr{L}$ to he the collection of leaves (regular and singular, which are limits of regular leaves) of the web structure. Each leaf has two complementary components. A *half space* is the closure of one of the complementary components . We let $\mathscr{H}$ denote the collection of half spaces. We let $\mathscr{U}$ denote the collection of DCC ultrafilters on $\mathbb{H}$. Technical point: we want ultrafilters defined on all but possibly finitely many halfspaces; when α is a principal ultrafilter defined by a point $x \in M$, we don't pick a side for leaves that contain x.

not sure if we want closure or not. details of proof will make it clear –ms

Properties of leafs that are needed:

1. Leaves are embedded
2. Any two leaves intersect in a leaf of the web structure of each leaf (each leaf inherits a local cubulation structure)
3. Three leaves that pairwise intersect have a nontrivial triple intersection (this ensures that the intersection pattern of leaves inside a leaf agrees with the intersection pattern of the leaves in the ambient space).

The proof follows what is done in dimension 2 (in top median diary).

Lemma 5.1. *The intersection of a nested collection of halfspaces is either empty or a halfspace.*

Proof. Import from diary and adjust for higher dimensions □

One constructs a map

$$\begin{aligned} M &\to \mathscr{U} \\ x &\mapsto \alpha_x \end{aligned}$$

where $\alpha_x = \{\mathfrak{h} \in \mathscr{H} | x \in \mathfrak{h}\}$. We now claim that this map is a bijection.

Lemma 5.2. *For each DCC ultrafilter $\alpha \in U$, there exists a unique $x \in M$ such that $\alpha = \alpha_x$.*

Proof. Start with a sequence of nested halfspaces in α, their intersection is again a halfspace, so we can apply zorn to obtain a halfspace $\mathfrak{h}_0 \in \alpha$ such that for all leaves $\ell \in \mathfrak{h}_0$, the halfspace chosen by α is the complementary halfspace of the halfspace contained in $\mathfrak{h}_0$ associated .

Now we are left to consider the remaining leaves for which we have not yet chosen a side. These are all the leaves that meet ℓ_0, the leaf associated to $\mathfrak{h}_0$. By induction the web structure on ℓ_0 has a median structure, so that a DCC ultrafilter on ℓ_0 determines a point on ℓ_0. Since leaves of the web structure of ℓ_0 are intersections with leaves in M, a DCC ultrafilter on ℓ_0 makes a decision for all the remaining leaves (except for finitely many). □

6 Surfaces

Web structure $\iff$ quartic differentials